



[Outline](#) > 2. Improving the model > 2.3 Reaction time (Euler's Method) > Calculation (Euler's method)

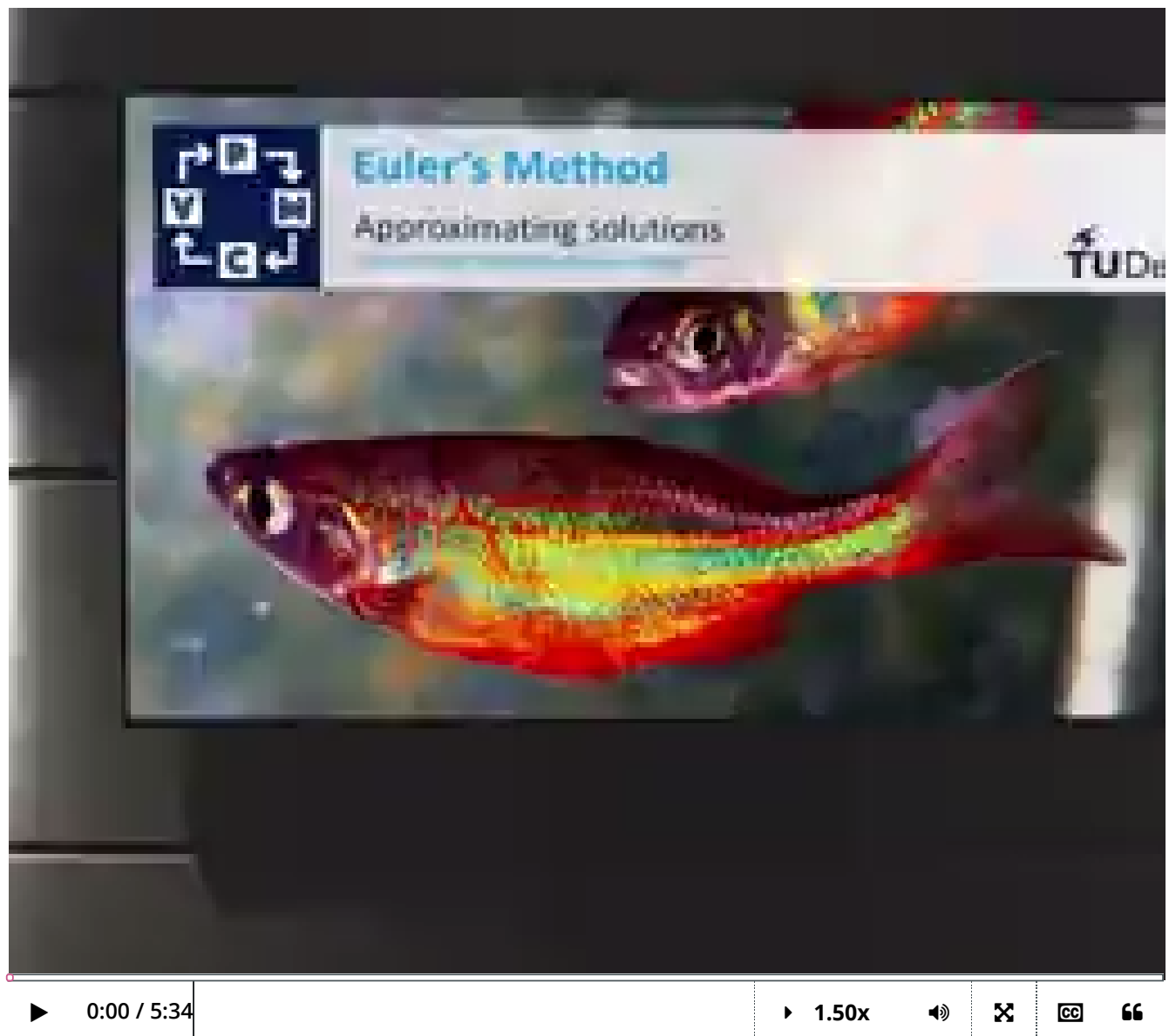
Calculation (Euler's method)

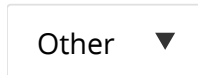
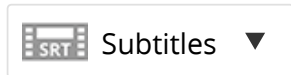
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The differential equation for the rainbowfish that we have now, could still be solved by hand. It would give you the analytical solution, which is exact. However, we are not doing that in this course, as we want to focus on the modelling, and not on analytical solution methods. We also do not want to limit our models to equations that can be solved analytically.

So, we will approximate the solutions of the differential equation with a numerical method, and we choose Euler's method. In the next video, this method is explained.

Video: Euler's method





Subtitles (captions) in other languages than provided can be viewed at YouTube. Select your language in the CC-button of YouTube.

In the video you have learned about Euler's method.

For the general differential equation

$$\frac{dy}{dt} = f(t, y),$$

the n -th step of Euler's method is given by

$$y((n+1)\Delta t) = y(n\Delta t) + \Delta t f(t, y(n\Delta t)),$$

in which Δt is some time step you have to choose.

In the following exercises, you are learning the method by doing a few of the steps "by hand", i.e. using a simple calculator. You will probably get bored with these calculations pretty soon, which is the purpose of the exercises :P. And then on the next page, you will use a computer program to automate these calculations.

The initial value problem to use:

$$\begin{cases} \frac{dP}{dt} = 0.7P \left(1 - \frac{P}{750}\right) - 20, \\ P(0) = 30. \end{cases}$$

The time step is set to one week, so $\Delta t = 7$ days.

Get out your calculator (app) and perform the first steps of Euler's method.

Exercise A

1/1 point (graded)

Given the initial condition $P(0) = 30$, what is the value of the right-hand-side function $f(t, P)$ at $t = 0$?

Please give your answer in two decimals.

✓ Answer: 0.16

0.16

You have used 1 of 3 attempts

i Answers are displayed within the problem

Exercise B

1/1 point (graded)

Given the answer above, we can now perform one time step with $\Delta t = 7$ days.The population size after one step, $P(7)$, is approximately

Please give your answer in two decimals.

✓ Answer: 31.12

You have used 1 of 3 attempts

i Answers are displayed within the problem

Exercise C

1/1 point (graded)

Given the answer above, we can now perform another time step with $\Delta t = 7$ days.The population size after two time steps, $P(14)$, is approximately

Please give your answer in two decimals.

✓ Answer: 37.28

You have used 1 of 3 attempts

i Answers are displayed within the problem

Exercise D

1/1 point (graded)

We now have done several steps for you and found $P(35) \approx 914.28$ and want to do another time step.The population size after another time step, $P(42)$, is approximately

Please give your answer in two decimals.

✓ Answer: -207.01

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You have used 1 of 3 attempts

 Answers are displayed within the problem

Exercise E

Question

What do you think of the value you calculated in the previous exercise?



Answer



Reflection



Results

Correct Answer

The value is correctly calculated, but not valid, because...

Although the number of rainbowfish is negative, the value is correctly calculated. A negative number of rainbowfish is not valid, as this is not realistic. So probably we need to investigate Euler's Method some more, which we will do in the next section.

Your initial answer: The value is correctly calculated, but not valid, because... (Option 2)

 "Population can't be negative. Using Euler method the answer was computed correctly, but time step is high that's why convergence not obtained."

Your final answer: The value is correctly calculated, but not valid, because... (Option 2)

 "Population can't be negative. Using Euler method the answer was computed correctly, but time step is high that's why convergence not obtained."

See how your classmates answered below

Class Breakdown

This is a look at how your classmates answered the question during the initial and final rounds.

Answer Options

Option 1

The value is correctly calculated and valid, because...

Option 2 (correct)

The value is correctly calculated, but not valid, because...

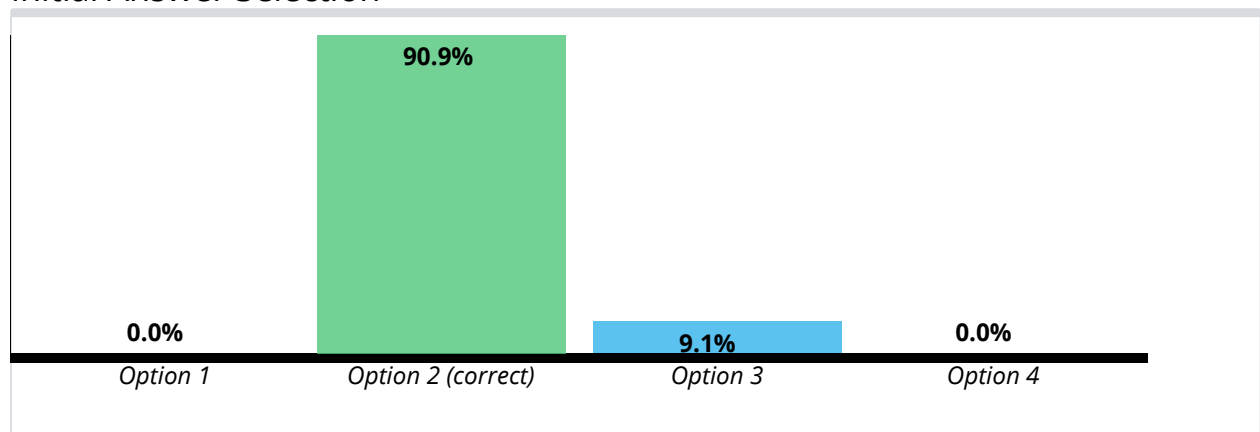
Option 3

The value is not correctly calculated, but valid, because...

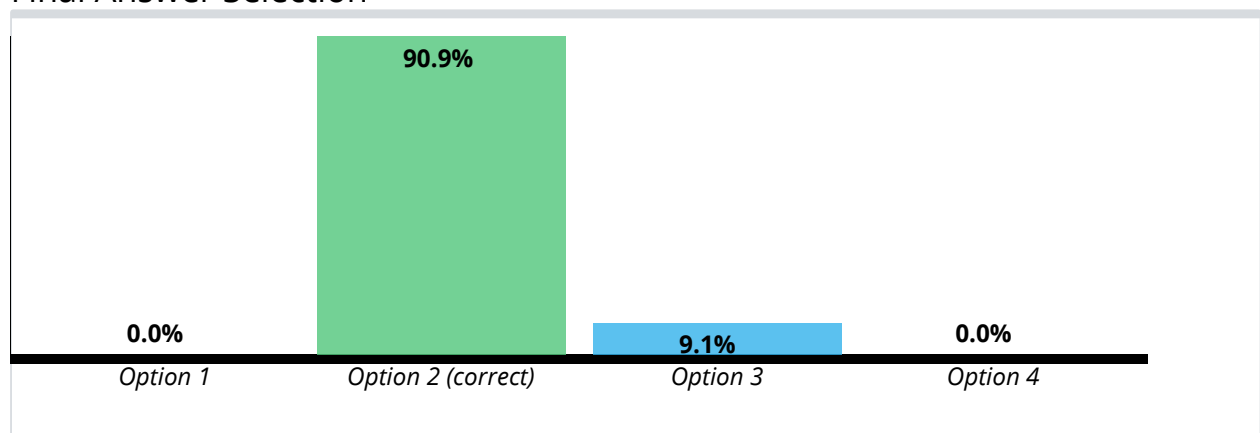
Option 4

The value is neither correctly calculated nor valid, because...

Initial Answer Selection



Final Answer Selection



As you might have noticed, calculating with Euler's Method by hand is a lot of work. Therefore we commonly do this by using software. In the next section you will learn how you can program Euler's Method yourself and investigate the weird result you encountered in this section.

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